

Data Structures

(CSE247)

Syllabus

Fall'25

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1 Basic Information

Credits	3 theory + 1 lab hours
Prerequisites	CSE-142 Object Oriented Programming Techniques

1.1 Objectives

This course surveys essential computer algorithms and data structures widely used in modern computing. It begins with fundamental data structures such as union-find, priority queues, binary search trees, balanced search trees, and hash tables, before progressing to advanced topics in graph algorithms and related applications.

The emphasis is on implementing these structures, analyzing their performance, and evaluating their effectiveness in solving practical computational problems.

2 Course Contents

- *Fundamental Data Structures*: Union-Find, priority queues, binary search trees, balanced search trees, hash tables.
- *Basic Analysis*: Asymptotic analysis, empirical measurements. Differences among best, average, and worst case behaviors of an algorithm. Complexity classes, such as constant, logarithmic, linear, quadratic, and exponential. Time and space trade-offs in algorithms.
- *Searching*: Binary search, symbol tables, binary search trees, balanced trees, hashing.
- *Graphs*: Undirected, directed, and weighted graphs; traversal and shortest path algorithms.

2.1 Weekly Schedule (tentative)

Table 1: Weekly schedule of topics and assessments

	Topic(s)	Notes
Week 1 <i>Aug 25 – Aug 31</i>	Union-Find Dynamic connectivity problem, union-find ADT. Quick find, quick union, weighted quick union, and path compression. Applications.	Homework 1 out
Week 2 <i>Sep 1 – Sep 7</i>	Analysis of Algorithms Empirical analysis. Mathematical model, running time. Big-O notation, properties of Big-O.	
Week 3 <i>Sep 8 – Sep 14</i>	Priority Queues Binary heaps, heap operations.	Homework 1 due Homework 2 out

Continued on next page

Table 1: Weekly schedule of topics and assessments (Continued)

	Topic(s)	Notes
Week 4 <i>Sep 15 – Sep 21</i>	Symbol tables Elementary implementations, binary search trees (BST).	
Week 5 <i>Sep 22 – Sep 28</i>	BST operations Insertion, deletion, search, traversal.	Homework 2 due Homework 3 out
Week 6 <i>Sep 29 – Oct 5</i>	Balanced BSTs 2-3 trees, red-black trees, Left-leaning red-black trees (LLRB).	
Week 7 <i>Oct 6 – Oct 12</i>	LLRB operations Insertion, deletion, search, traversal.	Homework 3 due
<i>Oct 13 – Oct 19</i>	Midterm Examinations	
<i>Oct 20 – Oct 26</i>	Semester Break	Homework 4 out
Week 8 <i>Oct 27 – Nov 2</i>	Hashing Hash functions, collisions, hash tables.	
Week 9 <i>Nov 3 – Nov 9</i>	Hashing (cont.) Separate chaining, linear probing, double hashing.	Homework 4 due Homework 5 out
Week 10 <i>Nov 10 – Nov 16</i>	Undirected Graphs Graph representation, graph API, depth-first search, connected components.	
Week 11 <i>Nov 17 – Nov 23</i>	Undirected Graphs (cont.) breadth-first search, shortest paths.	Homework 5 due Homework 6 out
Week 12 <i>Nov 24 – Nov 30</i>	Directed Graphs digraph API, digraph search, topological sort.	
Week 13 <i>Dec 1 – Dec 7</i>	Directed Graphs (cont.) strongly connected components.	Homework 6 due
Week 14 <i>Dec 8 – Dec 14</i>	Weighted Graphs edge-weighted digraphs, minimum spanning trees (Prim's, Kruskal's), Dijkstra's algorithm.	Final lab exam

2.2 References

Following are useful references:

- *Algorithms* by Robert Sedgewick and Kevin Wayne, Fourth edition, Addison Wesley, 2011.

3 Logistics

3.1 Staff and Office Hours



Taslim Murad

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Tabba Academic Block



Office hours: *Wed 9–12pm*



Imran Rauf

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Room 206, Tabba Academic Block



Office hours: *Tue/Thu 8:30am–11am*



Syed Ali Raza

Instructor

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Room 206, Tabba Academic Block



Office hours: *Mon 10–1pm*



Solat Jabeen Sheikh

Instructor

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Tabba Academic Block



Office hours: *Tue 3–5pm*



Mehwish Zafar

Instructor

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Tabba Academic Block



Office hours: *Tue 3–5pm*

3.2 Schedule

	<i>Lectures</i>	<i>Lab</i>
 99349	 <i>Ali Raza</i>  <i>MTC-23, Tabba Academic Block</i>  Mon/Wed: 8:30–9:45am	 <i>Ali Raza</i>  <i>MCL-3, Aman CED</i>  Wed: 10:00–12:45pm
 99351	 <i>Imran Rauf</i>  <i>MTC-25, Tabba Academic Block</i>  Mon/Wed: 8:30–9:45am	 <i>Mehwish Zafar</i>  <i>MCL-3, Aman CED</i>  Mon: 10:00–12:45pm
 99454	 <i>Imran Rauf</i>  <i>MTC-25, Tabba Academic Block</i>  Mon/Wed: 1–2:15pm	 <i>Imran Rauf</i>  <i>MCL-3, Aman CED</i>  Mon: 2:30–5:15pm
 99495	 <i>Solat Jabeen</i>  <i>MTC-25, Tabba Academic Block</i>  Tue/Thu: 10–11:15am	 <i>Mehwish Zafar</i>  <i>MCL-1, Aman CED</i>  Thu: 1:00–3:45pm
 99499	 <i>Taslim Murad</i>  <i>MTC-25, Tabba Academic Block</i>  Tue/Thu: 11:30–12:45am	 <i>Taslim Murad</i>  <i>MCL-1, Aman CED</i>  Tue: 1:00–3:45pm

3.3 Course Platforms

- **Learning Management System (LMS)**

<https://lms.iba.edu.pk/>

The LMS will be the primary platform for course-related activities, including assignment and lab submissions, distribution of course materials, announcements, and access to past exams/quizzes. Students are expected to check LMS regularly for updates.

- **University Management System (UMS)**

<https://ums.iba.edu.pk/>

UMS will be used for official course registration, attendance records, marks management, and final grade reporting. All official course records will be maintained on UMS.

- **Discord Server**

<https://discord.gg/YY44Bb4Gvj>

A dedicated Discord server will serve as the online discussion forum for this course. Students are encouraged to use it for asking questions, sharing ideas, and discussing homework problems (without posting full solutions). Instructors and TAs will monitor the server to provide guidance.

Discord Guidelines:

- Be respectful and professional in all communication.
- Do not post complete solutions to assignments, labs, or exams. You may discuss concepts, approaches, and debugging strategies.
- Use designated channels appropriately (e.g., homework, labs, general Q & A).
- Prefer posting questions in public channels instead of sending direct messages to instructors/TAs. This allows you to get faster feedback from peers and ensures others benefit from the discussion.
- Treat the forum as a collaborative learning space: help others when you can, but never do their work for them.
- Follow the IBA Code of Conduct and course academic integrity policy when participating.

4 Administration

4.1 Grading

The standard IBA grading scale will be used for this course. See <https://iba.edu.pk/gradingplan.php> for details. The final grade will be based on the following components:

Assessment	Weight
Assignments	15%
Quizzes	10%
Lab work	15%
Lab exam	10%
Midterm examination	20%
Final examination	30%

4.2 Academic Integrity & Code of Conduct

All students are expected to abide by the IBA Code of Conduct. Any form of scholastic dishonesty is a serious violation and will result in strict disciplinary action as per IBA regulations. Scholastic dishonesty includes, but is not limited to, plagiarism, cheating, and collusion.

- **Plagiarism:** Presenting someone else’s work, ideas, or expressions as your own without proper acknowledgment. This includes material that is copied, purchased, downloaded, or otherwise acquired. Submitting the same assignment/project for multiple courses without prior approval from all instructors also constitutes plagiarism.
- **Cheating:** Using or attempting to use unauthorized materials, information, or assistance in any academic exercise.
- **Collusion:** Assisting or attempting to assist another student in academic dishonesty, or failing to take reasonable precautions to prevent such assistance.

A second violation of academic integrity in this course will result in an automatic failing grade (F), in addition to other disciplinary actions that may be imposed under IBA policies.

4.3 Use of AI/LLMs

Tools such as ChatGPT, GitHub Copilot, and other large language models can be helpful when used responsibly, but they must not replace your own thinking and problem-solving.

4.3.1 Policy

- You may use LLMs only when you are stuck and need guidance or clarification.
- You must never copy/paste AI-generated text or code directly into your submission.
- Any work you submit must be written in your own words or coded by you.
- You must always be able to explain your solution; failure to do so will result in zero credit.

4.3.2 Why this policy?

Programming is best learned through the struggle of solving problems. Over-reliance on AI tools prevents you from building the problem-solving skills and habits required to succeed as a computer scientist. When used thoughtfully, AI can support your learning, but using it as a shortcut undermines your growth.

4.4 Learning Approach & Assessments

This course uses multiple assessment instruments—homework, quizzes, midterm, and final exams—all designed to reinforce your understanding of core concepts. True learning happens through the “personal struggle” of solving problems. Always attempt problems on your own before seeking hints or external help. Do not search for or rely on complete solutions online.

Collaboration on homework is permitted provided that:

- You write up your solutions individually.
- You cite all collaborators and sources clearly.
- Failure to follow these rules will result in zero credit

4.5 Late Work and Makeup Policy

- No late submissions will be accepted for assignments or labs.
- No makeup quizzes or exams will be administered.
- If you miss an assessment (lab, assignment, midterm) due to a valid medical excuse, the weight of the missed work will be shifted to the next scheduled exam.
- A valid excuse requires official documentation from a doctor.
- In the case of non-medical emergencies (e.g., family emergencies, university-approved events, or other exceptional circumstances), accommodations may be considered on a case-by-case basis, provided that timely documentation is submitted.
- Standard IBA policies will apply in all cases of conflicts.

4.6 Attendance Policy

IBA attendance policy applies.

5 Course Outcomes

The following table lists the course learning outcomes (CLO's) for this course.

<i>Course Learning Outcomes (CLO's)</i>	
CLO-1	Understand and Implement Core Data Structures:
	Students will be able to understand, design, and implement fundamental data structures such as union-find, priority queues, binary search trees, balanced search trees, hash tables, and graphs.
CLO-2	Analyze Algorithm Efficiency:
	Students will be able to analyze the time and space complexity of algorithms using Big-O notation, and compare the performance of different algorithms for tasks such as dynamic connectivity, searching, and graph processing to select the most appropriate approach for a given problem.
CLO-3	Apply Data Structures in Problem Solving:
	Students will be able to apply appropriate data structures and algorithms to design and implement efficient solutions for computational problems, and evaluate their effectiveness in practical applications.

The mapping of course learning outcomes (CLO's) to program learning outcomes (PLO's) is as follows:

	Academic Education	Knowledge for Solving computing Problems	Problem Analysis	Design / Development of Solutions	Modern Tool Usage	Individual and Team Work	Communication	Computing Professionalism and Society	Ethics	Life-long Learning
	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO-10
CLO-1		✓								
CLO-2			✓							
CLO-3				✓						

The following table identifies the assessment instruments which will be used to assess the CLO's.

	CLO-1	CLO-2	CLO-3
	Core Data Structures	Analyzing algorithms	Problem solving
Labs		✓	
Homeworks		✓	✓
Quizzes	✓		
Exams	✓		✓

For reference, the graduate attributes (program learning outcomes - PLO's) taken from <https://www.seoulaccord.org/document.php?id=79> are as follows.

- **PLO-1. Academic Education**

[Educational depth and breadth]

Completion of an accredited program of study designed to prepare graduates as computing professionals

- **PLO-2. Knowledge for Solving Computing Problems**

[Breadth and depth of education and type of knowledge, both theoretical and practical]

Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements

- **PLO-3. Problem Analysis**

[Complexity of analysis]

Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines

- **PLO-4. Design / Development of Solutions**

[Breadth and uniqueness of computing problems, i.e., the extent to which problems are original and to which solutions have previously been identified or codified]

Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

- **PLO-5. Modern Tool Usage**

[Level and appropriateness of the tool to the type of activities performed]

Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations

- **PLO-6. Individual and Team Work**

[Role in, and diversity of, the team]

Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings

- **PLO-7. Communication**

[Level of communication according to type of activities performed]

Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions

- **PLO-8. Computing Professionalism and Society**

[No differentiation in this characteristic except level of practice]

Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice

- **PLO-9. Ethics**

[No differentiation in this characteristic except level of practice]

Understand and commit to professional ethics, responsibilities, and norms of professional computing practice

- **PLO-10. Life-long Learning**

[No differentiation in this characteristic except level of practice]

Recognize the need, and have the ability, to engage in independent learning for continual

development as a computing professional